Fail-Closed or Fail-Degraded? A Simulation Study of Forced Assurance Loss in Contested Multi-Hop QKD Networks

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***Abstract*—Stopping quantum key generation need not immediately exhaust a trusted-relay network's stored keys. We study how finite reserves, interruption duration and fallback policy jointly determine timely key service and assurance loss. Generation, transport and online key-establishment failures are separated, with paired policy comparisons within each condition. A 360-run, five-topology grid includes 24-hour generation cuts with only 2% of usable capacity initially filled. In the eight-node formation, fail-closed service is 6.70%, compared with 97.53% under fallback; compliant fallback service is only 6.61%. The fraction of offered requests entitled to information-theoretically secure (ITS) key delivery but served below it is 90.08% (95% interval 89.20–90.97%). With full starting reserves, the same cut produces no observed forced loss in the 24-hour cohort. An interpretable controller trained across several conditions raises mean count-based utility by 0.0011 on a held-out benchmark, yet increases associated below-ITS payload exposure by 1.71 percentage points while respecting every class exception cap. The results characterize modeled key service and contractual exposure, rather than payload throughput or field-validated security.**

***Index Terms*—Quantum key distribution, trusted relays, key management, post-quantum cryptography, assurance metrics, constrained policy search.**

I. Introduction

Quantum key distribution (QKD) supply interruptions can be handled by waiting, blocking or computational fallback. ITU-T Y.3803 permits policy-controlled symmetric relay substitution and records the method [1]; Tysowski *et al.* describe fallback and blocking [2], and Spooren *et al.* implement layered failure timers [3]. We examine how these choices interact with usable stored key.

A relay formation may mix command-key requests, an expensive one-time-pad (OTP) class and logistics traffic. We distinguish successful key service from satisfaction of its assurance requirement, and loss of replenishment from loss of accepted stored key.

We evaluate finite reserves, interruption duration, relative link-suppression budgets and learned grade allocation under class-specific exception permissions. Timely and floor-compliant service share an arrival denominator; entitlement to information-theoretically secure (ITS) keys, payload and class-importance metrics expose different losses. A cumulative cut-budget inequality and exact fixtures delimit feasible service. The learning study compares a compact controller with a greedy policy under the same monitor.

II. Related Work

Adaptive graded key delivery already appears in QuLore [4], and Q-learning has been applied to resource-aware trusted-relay routing [5]. Our learned action is a requested assurance grade. Routing uses the explicitly defined implementation below; the study does not compare against reimplemented published routing algorithms.

Opportunistic QKD studies single-link reservoirs and correlated depletion/recovery [6]; QuIKS combines end-to-end buffering, reactive control and device traces [7]; Zhu studies finite buffers, fallback fractions and class-sensitive output [8]. Building on these, we jointly evaluate targeted generation interruptions, relative link costs, paired excess entitlement loss and learned class-exception allocation.

Established runtime filtering [9] motivates our deterministic prefix-count exception monitor, whose contract is specified below.

III. Model and Assurance Accounting

A. Stored key and failure surfaces

A trusted-relay graph has one shared key store per undirected link. Relaying a session key debits every traversed link once. For link e, stored stock obeys

$$B_e(t) = B_e(0) + G_e(t) - C_e(t) - O_e(t) - P_e(t), \quad (1)$$

where G_e is gross incoming distilled-key supply before buffer overflow, C_e useful relay consumption, O_e overflow discard, and P_e intentional working-key purge. Usable stock is $U_e = (B_e - b_{\min})_+$. Admission checks every hop before any debit; partial relay transactions and in-flight failures are outside this abstraction.

Stored keys are retained while synchronized, unexpired and uncompromised. The purge alternative discards spendable stock when generation fails but preserves the protected reserve. These alternatives reflect the generation/storage distinction [1] and implementation-dependent shutdown or synchronization policies [2].

Generation suppression leaves classical relay transport available. A transport-disconnected request cannot be served, even

with pre-positioned key material. An isolated online key-establishment failure is a third surface: it can prevent a key-encapsulation-mechanism (KEM) fallback without blocking a valid pre-positioned symmetric key. These surfaces are modeled separately.

Nominal generation is a parameterized input. For quantum bit error rate (QBER) x, nominal $x_0 = 0.03$ and reconciliation factor $f = 1.16$, we use the reduced response

$$\frac{R_e(x)}{R_{e,0}} = \left[\frac{\max\{0, 1-(1+f)H_2(x)\}}{1-(1+f)H_2(x_0)}\right]^{\gamma} \tag{2}$$

below x_{abort}, and zero at or above it; H_2 is binary entropy in bits and the ratio is capped at one. The engineering threshold is 0.075, with 0.11 as a sensitivity. The entropy numerator can become zero before a selected abort threshold, so the latter alone does not set the zero-rate boundary. The exponents 1 and 3.4 are illustrative response shapes, not a cross-modality calibration or a finite-key security proof.

B. Grades, requests and debits

The policy order is G1, G2, G3, G4. G1 requires payload-length OTP key; G2 a 256-bit QKD-delivered symmetric key; G3 128 QKD bits as a hybrid ingredient; and G4 no quantum key. G2 classifies information-theoretic key delivery with computational payload encryption. The ordering specifies contracts, not universal cryptographic strength or hybrid-combiner security.

For a nonzero quantum-key demand D_g, the per-hop debit is

$$q_g = D_g + 240 + 127 n_{\text{auth}} \quad \text{bits}. \tag{3}$$

The metadata allocation is an explicit choice within the key-file abstraction of Y.3803 [1]. The tag size is motivated by Walenta *et al.* [10]; reusable-hash authentication charges are discussed by Fung *et al.* [11]. These sources do not establish a universal number of authenticated messages per relayed request. We use $n_{\text{auth}} = 7$ as a conservative relay-overhead setting relative to the optimistic value 2, and also evaluate 40. The corresponding G2 debits are 1385, 750 and 5576 bits per hop. This is a simplified cost model, not protocol-level calibration.

Command-and-control (C2), intelligence/surveillance/reconnaissance (ISR), and logistics (LOG) requests have payloads 2048, 144000 and 320000 bits; floors G2, G1 and G3; deadlines 5, 30 and 120 seconds; and strict scheduling priorities in that order. Independent Poisson arrival rates are 0.05, 0.002 and 0.03 per second in the reduced-load case. These are declared workload scenarios. The required mean key debit is 397.218 bits/s across a cut traversed once per request, compared with 480 bits/s for two nominal 240-bit/s branches. This is a nominal capacity check, not a queue-stability proof. ISR arrival-rate factors 0.03, 0.1 and 1 relative to 0.02/s separate lighter load from overload.

C. Common-denominator outcomes

For all N requests arriving in a fixed observation window, let S receive key service within their deadlines and $D \subseteq S$ receive a grade below their class floor. Define

$$A = \frac{|S|}{N}, \qquad Q = \frac{|S \setminus D|}{N}, \qquad L = \frac{|D|}{N}. \tag{4}$$

Thus $A = Q + L$; blocking is $1 - A$. This identity makes fallback's service advantage distinguishable from floor compliance.

Let E be requests entitled to G1 or G2, and $D_{\text{ITS}} \subseteq E \cap S$ those served at G3 or G4. Offered-request forced assurance loss is $F = |D_{\text{ITS}}|/|E|$. ITS-delivered, below-ITS-delivered and blocked fractions sum to one on this entitled-arrival cohort. The accompanying tables retain $F_{|S} = |D_{\text{ITS}}|/|E \cap S|$ separately. LOG is excluded from the ITS-entitled denominator.

For associated payload size b_s, define $F_B = \sum_{s \in D_{\text{ITS}}} b_s / \sum_{s \in E} b_s$; L_B similarly weights below-floor service over all arrivals. These are exposure proxies, not observed compromise or calibrated harm. Class-importance vectors $(1,1,1)$, $(5,2,1)$ and $(10,3,1)$ provide another transparent reweighting. Attack-attributable change is a paired attacked-minus-unattacked difference under the same exogenous environment; no zero baseline is assumed.

D. A finite cut-budget check

For an edge cut $\delta(W)$, $W \subset V$, every compliant relay between separated endpoints must consume key on a crossing edge. Let $\mathcal{C}_T$ contain these compliant requests completed by time T, and q_s be their minimum per-hop debit. Then

$$\sum_{s \in \mathcal{C}_T} q_s \le \sum_{e \in \delta(W)} \left[U_e(0) + G_e(T)\right]. \tag{5}$$

Discard, competing traffic and repeated cut crossings only tighten this upper bound. It follows by summing conservation over crossing stores. A generation cut removes replenishment, not the initial term.

The pooled bound need not be attainable. With three disjoint two-hop branches, 1900 spendable bits per edge and a 750-bit debit, it permits seven requests, but the simulator serves only six: each branch retains an unusable 400-bit fragment. Four 749-bit stores permit three requests under the pooled relaxation and none in the actual graph. Exact divisible-stock and purge cases also pass. These fixtures check implementation and the bound's scope.

IV. Policies and Experimental Protocol

A. Compared control policies

Fail closed waits for the authorized floor and blocks at deadline. Unrestricted fallback attempts progressively weaker grades when that floor cannot be met. Our shortest-path baseline chooses the first transport-usable minimum-hop candidate. The key-aware implementation minimizes bottleneck estimated refill time plus a 0.5-second penalty per hop. Candidate enumeration is limited to four simple paths within two

extra hops of the shortest. The comparison concerns these implementations.

The learned controller uses 18 states: three classes, three floor-feasibility categories, and two deadline-urgency categories. Feasibility distinguishes affordable now, refill estimated within the remaining deadline, and neither; urgency means at most 20% of the deadline remains. Estimated refill uses current rates and ignores future competing arrivals. The five proposed actions are the class floor, strongest affordable fallback, defer, G4 and G2.

All learned and budget-greedy actions pass through the same monitor. If $N_c(t)$ class-c requests have arrived and $V_c(t)$ below-floor services have been committed, a proposed exception is permitted only when

$$V_c(t) + 1 \leq \lfloor \epsilon_c N_c(t) \rfloor, \tag{6}$$

with $(\epsilon_{\mathrm{C2}}, \epsilon_{\mathrm{ISR}}, \epsilon_{\mathrm{LOG}}) = (0, 0.1, 0.1)$. Failed attempts spend no allowance. Since arrivals only increase the right-hand side and every committed exception is checked before use, induction gives the prefix bound. Completed exceptions are a subset of committed exceptions. This is an arrival-count permission guarantee within the simulator, not a learned cryptographic guarantee. Strict-floor and greedy policies are explicit baselines.

Categorical cross-entropy policy search [12] uses ten unique tables per generation, four generations and two optimizer restarts. Every generation includes the strict, greedy and incumbent tables. Short-nominal training uses six-hour rollouts; full-nominal uses 24 hours; full-robust uses a mean over nominal, generation-cut and correlated low-rate conditions. Each candidate receives six rollouts. Validation selects between the two training-selected finalists on three conditions and four separate seeds. Final testing uses twelve previously unused seeds over seven fixed conditions, including unseen topologies. The objective $J = A - 0.75L = Q + 0.25L$ assigns one-quarter value to below-floor service. Tables, histories, objective, validation rule and test plan are frozen before final testing; optimality is not established.

B. Scenarios, uncertainty and verification

The graph families are an eight-node, nine-link formation; an eight-node ring; a 3×3 grid; and three or four parallel three-hop branches. The latter have $(8, 9)$ and $(10, 12)$ nodes/edges. Individual formation terminals are leaves, with cut size one; isolating all formation terminals requires two edges. The other source-terminal cut sizes are two, two, three and four. These different notions of cut are not conflated.

Formation rates are 200–400 bits/s on eight free-space links and 300 kbits/s on one fiber; the other graphs use 240-bit/s links. Capacities, threshold and reserve are selected legacy QKDNetSim example parameters [13]: 52428800, 11324620 and 1048576 bits. Capacity is 50 Mibit, not a measured hardware specification. Our overflow model clamps continuous stock and discards excess; it is not byte-for-byte equivalence to a discrete block buffer. Initial spendable fractions are 0.02, 0.1 and 1; the main stressed initial stock is 1027604.48 usable bits per edge.

TABLE I
FORMATION CLASS OUTCOMES UNDER THE TWO-EDGE GENERATION CUT. ENTRIES ARE PERCENT OF CLASS ARRIVALS; 95% INTERVALS APPEAR BELOW THE MEANS, FROM 12 SEEDS.

Class	Closed Q	Fallback A	Fallback Q
C2	6.67 (5.70, 7.61)	96.72 (96.52, 96.89)	6.65 (5.68, 7.59)
ISR	5.67 (5.41, 5.97)	97.50 (96.66, 98.32)	5.67 (5.41, 5.97)
LOG	6.83 (5.91, 7.67)	98.88 (98.74, 99.01)	6.60 (5.73, 7.41)

Link-suppression cost is one for free-space links and ten for fiber, with ratios one and one hundred as sensitivities. These are relative scenario weights, not measured adversary expenditure. A weighted source-to-all-terminals cut is a structural reference, not an optimal finite-horizon attack. Independent free-space generation and all-link transport failures use declared mean up/down times of 3600/120 seconds. The correlated-generation sensitivity gives all free-space generators a common 900/600-second up/down process. Policy comparisons within each condition share exogenous draws that are independent of policy and attack choices.

Simulation uses one-second steps and a 24-hour arrival window without a steady-state claim. Pending requests are followed to their deadlines; an interruption scheduled to end at 24 hours can release before the last completion. Equation 5 uses the completion horizon for such a cohort. Twelve seeds support the main grid, eight the sensitivity and fallback comparisons, and six the threshold sweeps. Percentages are means of seedwise proportions. Intervals are marginal 95% percentile bootstrap intervals over independent seeds, with 20000 resamples, conditional on the parameters and, for learning, the selected table.

Verification checks conservation, reserve and deadline invariants, exact resource fixtures, matched exogenous traces and prefix budgets. Published sources motivate parameter choices; they do not independently validate the network predictions or cryptographic security.

V. RESULTS

A. Availability is not compliant service

At the main low-reserve setting, suppressing the weighted generation cut leaves formation fail-closed service at 6.70%, compared with 97.53% without attack. Fallback serves 97.53% of arrivals, but only 6.61% meet their original floor. Its compliant-service difference from fail closed is -0.09 percentage points, with interval $[-0.11, -0.07]$. The large availability difference therefore cannot be interpreted as preserved authorized protection.

Figure 1 shows the same distinction across all five graphs. Table I prevents an aggregate score from hiding a class's admission failure. Formation forced loss is 90.08% of offered

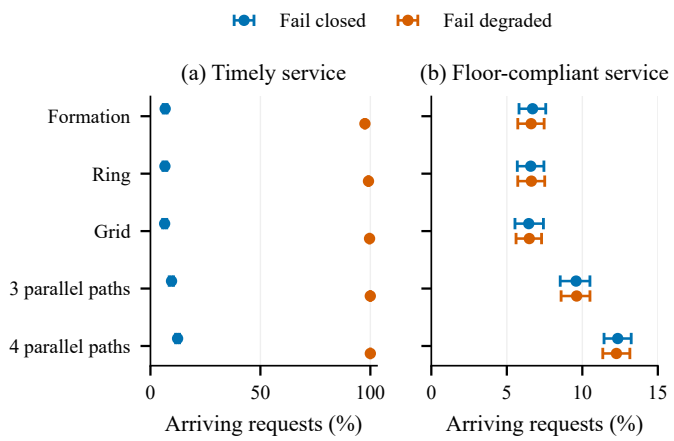


Fig. 1. Timely and floor-compliant key service under full generation-cut suppression, 24-hour arrival cohorts and 2% initial usable stock. Bars are 95% seed-bootstrap intervals ($n = 12$). The horizontal scales differ.

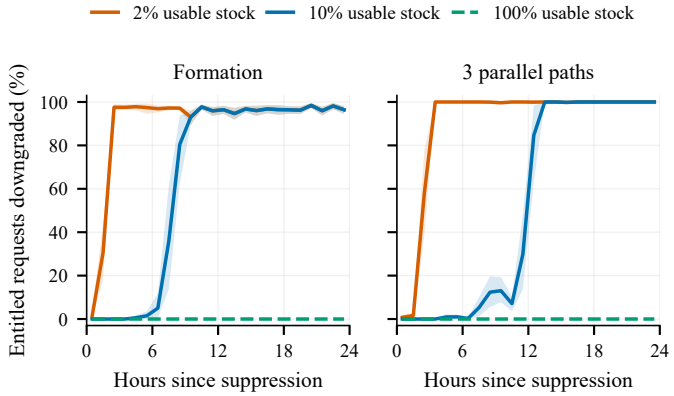


Fig. 2. Hourly entitled-arrival loss under a held generation cut, with 2%, 10% and 100% initial usable fill. Lines and pointwise 95% bands use eight seeds. These are hourly cohorts, not a cumulative failure probability.

ITS-entitled requests, whereas below-floor payload exposure L_B is 92.24% of all associated payload bits. The different values reflect different denominators and requirements.

B. Reserve and operating-envelope dependence

The eight-seed stock sweep changes the interpretation substantially. Under the same formation cut, fail-closed service is 6.97%, 31.65% and 97.54% at 2%, 10% and 100% initial usable fill. Full-stock runs show no forced loss and the two modes have equal service. At 2% fill, six-hour suppression produces $F = 17.62\%$ $[17.07, 18.16]$, compared with 89.85% for 24 hours. Figure 2 shows why a cut size without a stock level and observation window is insufficient.

Accounting remains consequential: raising n_{auth} from 2 to 40 reduces closed service from 8.53% to 3.27%, a paired loss of 5.26 points $[4.34, 6.07]$. Common-cause generation outages already produce unattacked $F = 2.96\%$ $[1.57, 4.39]$; the paired attack excess is 86.89 points $[85.57, 88.21]$. Observed loss is therefore not automatically attack-attributable.

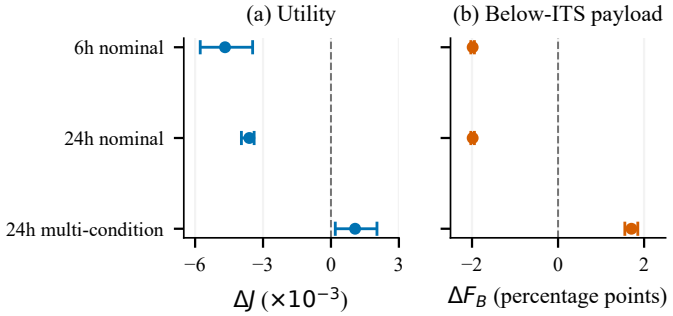


Fig. 3. Learned policies relative to budget-greedy. Seven condition means are first averaged within each of 12 held-out seeds; bars are paired 95% intervals. Higher J is favorable, while higher F_B means more associated entitled payload served below ITS.

With $\gamma = 1$, no environmental outages and increment $\Delta x = 0.045$, full three- and four-path cuts produce $F = 90.17\%$ $[89.47, 90.88]$ and 87.47% $[86.40, 88.71]$; at increment 0.03 both are zero. At $\Delta x = 0.04$, absolute QBER is 0.07, below the 0.075 engineering abort. The three-path case then gives $F = 0$ for $\gamma = 1$ and 83.47% $[82.67, 84.34]$ for the illustrative $\gamma = 3.4$ response. Slow positive generation can force loss before abort; an engineering threshold and a minimum cut do not define a universal finite-horizon forcing rule.

C. Target persistence and attack cost

At tested two-path chain depths $d = 2, \ldots, 6$, deterministic fixed-centrality prefixes require $d + 1$ or $d + 2$ removals depending on tie handling, despite connectivity two. At depths two, four and six, we also test an online selector using past committed-route counts and current transport reachability. Every 60 seconds it maximizes uncovered route weight per link cost; unobserved viable paths receive weight 0.01. A six-seed exploratory study releases targets when routes become unusable. A post-hoc diagnostic on six disjoint seeds retains targets to fill unused slots.

At depth six and two slots, retaining targets raises excess F from 35.50% to 89.69%, a paired increase of 54.19 points $[49.72, 58.08]$. The persistent variant spends 48 relative-cost link-hours and becomes a fixed cover, matching the structural-cut outcomes at depths two, four and six. This identifies persistence/exposure as the intervention, not an intrinsic advantage of adaptivity. Concurrent capacity, realized link-time and distinct links touched are reported separately; retargeting overhead is unmodeled.

D. Learning and the exposure trade-off

Across the seven fixed test conditions, the multi-condition controller improves mean utility over budget-greedy by $\Delta J = 0.0011$ with interval $[0.0002, 0.0020]$. Mean compliant service increases 0.10 percentage points. This is a small, objective-dependent gain: associated below-ITS payload exposure rises from 1.99% to 3.69%, a paired increase of 1.71 points $[1.55, 1.85]$.

The class breakdown explains the aggregate: C2 and LOG compliant service increase 0.13 and 0.25 points, while ISR

compliant service falls 2.77 points. All recorded prefix audits respect the declared class caps. The two additional priority vectors preserve the small positive mean utility difference when reweighting these frozen outcomes. The six-hour and 24-hour nominal training regimes instead change mean utility by -0.0047 and -0.0036 relative to greedy; the latter selects the strict-floor table. Figure 3 reports these unfavorable outcomes alongside the multi-condition gain. Count-based scores and class exception caps therefore leave a separate payload-exposure trade-off.

E. Key establishment is not data transport

With the generation cut and online key-establishment service unavailable, timely key service is 5.03%. Ideal pre-positioned secrets restore 97.54%, a paired gain of 92.51 points $[91.91, 93.08]$, but compliant service improves only 1.90 points $[1.15, 2.62]$. Under the transport cut both variants serve 0.064% of the arrival cohort, entirely after cut release during deadline follow-up. Neither serves through the active cut. This pre-positioned-key comparison assumes sufficient unexpired material; lifetime, refresh and distribution costs are outside the model.

VI. Discussion and Limitations

The simulator assumes trusted relays, synchronized valid stores, instant state information and atomic debits. It does not model payload transmission, cipher/KEM execution or compromised key material. Finite candidate routes, fixed class payloads and illustrative supply/outage processes limit external validity. Payload and priority weights summarize modeled exposure, not realized harm.

Finite buffers, depletion analysis and runtime shielding have established precedents. The practical question here is their interaction with explicit entitlement, interruption cost and a constrained learned controller. A favorable scalar utility does not waive a class requirement, and a low forced-loss value can result from blocking rather than preserving service. Both outcomes must remain visible. Training-horizon comparisons concern the registered protocols, including their different simulated exposure and state coverage; they do not isolate depletion as the unique cause of a learned-policy difference.

VII. Conclusion

For finite-reservoir trusted-relay QKD, a generation cut, a key-store policy and a service policy are separate decisions. Starting stock, rate response and target persistence prevent cut size alone from determining finite-horizon assurance loss. Fallback can preserve timely key service while adding little compliant service. A learned allocator can also improve a count-based score within all class caps while increasing entitled-payload exposure. Reporting these outcomes together makes the operational trade-off explicit.

Reproducibility and Acknowledgement

The artifact includes fixed configurations, raw per-seed outcomes, source hashes, numerical-table and figure generators, tests, policy-search histories and a validation-selected policy freeze. Historical outputs are preserved separately. The authors used a large language model to assist coding, analysis and drafting; the reported numerical evidence is generated by the accompanying executable experiments.

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